RTDL: A Python-Hosted Ray-Tracing DSL with Bounded Reproduction of the RayJoin Workload Family

Anonymous Submission

Abstract—RTDL (Ray-Tracing Domain Language) is a Python-hosted domain-specific language and compiler/runtime stack for non-graphical ray-tracing workloads. This paper presents the current RTDL solution through the RayJoin workload family as a bounded, externally checked, multi-backend vertical slice. The system combines a Python authoring surface, a backend-independent compiled-kernel representation, a native C/C++ oracle, and controlled execution backends for Intel Embree and NVIDIA OptiX, with Vulkan retained as a provisional backend. We evaluate RTDL against the workload and artifact structure of the RayJoin paper using accepted bounded packages on a Linux host and an external PostGIS reference. The resulting package closes the tractable RayJoin experiment surface under an explicit bounded/analogue rule: available families are validated across PostGIS, the native oracle, Embree, and OptiX; synthetic scalability figures are reproduced as bounded analogues; and dataset families with unstable or unavailable public acquisition paths are deferred explicitly rather than overclaimed. The paper reports the RTDL design, the evaluation methodology, the resulting correctness and performance evidence, and the precise boundaries that separate bounded closure from paper-identical reproduction.

Index Terms—domain-specific language, ray tracing, spatial joins, Embree, OptiX, PostGIS, RayJoin, reproducibility

I. Introduction

Ray-tracing hardware and software stacks are increasingly useful outside graphics, but programming them directly still imposes a heavy systems burden on application authors. Spatial join workloads are a clear example: the algorithmic idea may be natural, yet the actual implementation often requires direct management of acceleration structures, ray payloads, launch parameters, runtime plumbing, and numerical boundary behavior. RTDL addresses that problem by raising the programming surface to a compact kernel language while preserving a backend path capable of executing RayJoin-style workloads across CPU and GPU targets.

The current RTDL project is broader than a single application, but RayJoin is its first serious target because the workload family is large enough to stress the language/runtime boundary and structured enough to admit rigorous external checks. This paper therefore does not present RTDL as a finished general system. Instead, it presents a bounded, trustworthy vertical slice: a Python-hosted language, a compiler/runtime pipeline, and a multi-backend evaluation package that mirrors the RayJoin paper structure as far as current stable data and infrastructure allow.

The main contribution of this paper is not a claim of paper-identical reproduction. Rather, the contribution is a bounded reproduction package with explicit fidelity labels and external correctness checks. RTDL now supports accepted bounded closure for representative RayJoin-style workloads across a native C/C++ oracle, Embree, OptiX, and PostGIS on selected real-data packages. It also supports bounded analogues of the RayJoin scalability and overlay surfaces. Dataset families whose public acquisition paths are unstable or unavailable are deferred explicitly.

In summary, this paper makes four contributions:

1) It describes RTDL as a Python-hosted language and compiler/runtime system for non-graphical ray-tracing applications, using RayJoin as the current validation slice.
2) It documents a trusted execution stack consisting of a native semantic oracle, controlled Embree and OptiX runtimes, and a PostGIS external checker.
3) It presents a bounded RayJoin experiment package covering LSI, PIP, and overlay-seed analogue workloads across accepted available families.
4) It states precisely what has been reproduced, what is only an analogue, and what remains deferred because of unavailable public data.

II. Background and Problem Setting

RayJoin demonstrates that ray-tracing techniques can accelerate spatial joins, but it also exposes the implementation burden of direct backend programming. RTDL is motivated by the same application class while aiming to decouple kernel semantics from backend-specific realization. In the current repo, the whole project vision includes multiple backend families, but the validated slice is intentionally narrower: RayJoin-style workloads executed through a native oracle, Embree, and OptiX, with Vulkan retained only as a provisional path.

The evaluation surface mirrored from RayJoin includes three workload families that matter directly for the current paper:

- LSI: line-segment intersection style joins,
- PIP: point-in-polygon style joins,
- Overlay: currently represented in RTDL as an overlay-seed analogue, not full polygon overlay materialization.

For this work, RayJoin Figure 13 and Figure 14 are represented as bounded synthetic analogues, while RayJoin Table 3, Table 4, and Figure 15 are represented through accepted bounded packages on stable and available dataset families.

III. Related Work

A. Spatial Join Systems

RayJoin is the closest direct comparison point for this paper because RTDL is explicitly evaluated against its workload family, artifact structure, and reporting surfaces [1]. PostGIS remains the most important external reference in the current RTDL evaluation because it provides a widely used industrial spatial database with indexed spatial predicates and well-understood geometry semantics [2]. GeoSpark SQL represents a different design point: it exposes SQL-level spatial operations on top of a distributed Spark/Hive execution substrate rather than a backend-portable kernel language [3]. Other widely used distributed spatial systems, such as SpatialHadoop and Apache Sedona, occupy the same broad system class even though they expose different storage and execution tradeoffs [4], [5]. RTDL differs from all of these classes of systems. It is not a database, and it is not a paper-specific one-off implementation. Instead, it aims to express non-graphical ray-tracing workloads through a constrained kernel language and then lower them into multiple execution backends while keeping explicit correctness contracts.

B. Compiler and DSL Context

RTDL also relates to compiler-backed domain-specific systems such as Halide [6] and TVM [7]. Those systems separate high-level computation intent from lower-level schedule or backend realization and are therefore important precedents for RTDL's own authored-kernel to backend-plan split. The difference is domain focus. Halide and TVM center image-processing and tensor-operator pipelines, whereas RTDL centers geometric candidate generation, refine predicates, and emitted join rows. The relevant design pressure is therefore not only optimization, but exact spatial semantics across heterogeneous backends.

C. Ray-Tracing Runtime Context

At the runtime layer, RTDL builds on ray-tracing infrastructure rather than reimplementing traversal from scratch. Embree provides the validated CPU ray-tracing kernel framework [8], while OptiX provides the accepted GPU execution path in this paper [9]. RTDL's contribution is not a new traversal engine. The contribution is the language/runtime boundary, the native oracle, and the bounded multi-system evidence package built on top of these runtimes.

IV. RTDL Design

A. Language Surface

RTDL is authored from Python through a constrained kernel DSL. A user writes a kernel by declaring typed inputs, traversal intent, refine predicates, and output fields. The intent is that authors specify workload meaning rather than backend launch details. The same language surface currently supports six implemented workloads in the repository, but the RayJoin-facing slice of this paper focuses on LSI, PIP, and overlay.

B. Compiled Representation and Lowering

RTDL lowers authored kernels into a compiled-kernel representation that separates language semantics from backend realization. The lowering path remains RayJoin-oriented in the current codebase: it maps geometry roles, candidate generation, refine predicates, and emitted rows into backend-specific execution plans. This allows the same kernel to execute against a semantic oracle, a CPU Embree runtime, and a GPU OptiX runtime.

C. Execution Backends

The accepted backends for this paper are:

- Native C/C++ oracle: the current semantic ground truth for accepted results.
- Embree: the strongest validated CPU backend in the repo.
- OptiX: the bounded, controlled GPU backend validated on the Linux workstation in Table I.

The Vulkan backend remains in the repository but is explicitly excluded from the accepted paper package because it is still provisional.

V. Methodology

A. Correctness Method

The correctness story is intentionally layered. The native C/C++ oracle is the internal ground truth. Embree and OptiX are checked against that oracle. For accepted bounded real-data packages, PostGIS provides an external industrial reference. PostGIS queries are executed with indexed GiST-assisted plans rather than brute-force SQL. For PIP, PostGIS positive-hit queries are expanded into the same full-matrix RTDL truth semantics before parity comparison.

The accepted PIP parity path is also more specific than a generic manual point-in-polygon implementation. After the Goal 50 external comparison exposed remaining boundary/topology mismatches, the accepted RTDL refine path adopted GEOS-backed prepared-polygon covers(point) semantics for the native oracle and Embree, and the accepted OptiX overlay result was observed on a GEOS-enabled build on the validated Linux host [10]. This detail matters because the paper claims exact bounded parity with PostGIS on the accepted packages, not merely approximate geometric agreement.

TABLE I
Execution environment for the accepted bounded four-system package.

Item	Value
Host class	Linux validation workstation
OS	Ubuntu 24.04.4 LTS
CPU	Intel Core i7-7700HQ
Memory	about 15 GiB
GPU	NVIDIA GeForce GTX 1070
Driver	580.126.09
CPU RT backend	Embree 4.3.0
GPU RT backend	OptiX 9.0.0
External checker	local PostgreSQL/PostGIS on the same host
GEOS role	exact prepared-polygon covers(point) path for accepted PIP parity

Parity is checked at the emitted-row level. In the accepted underlying artifacts, each system produces a canonical row order, and the final row stream is hashed with SHA-256. This paper therefore includes only cases whose final accepted artifacts already established exact hash-level agreement across the compared systems.

B. Evidence Policy

This paper uses only accepted artifacts from the repository's audited goal history. A case is counted here only if it is already accepted and honestly scoped. The paper therefore distinguishes:

- done-bounded: accepted bounded analogue or bounded package,
- deferred-unavailable: excluded because the public acquisition path is unstable or unavailable,
- not claimed: paper-identical or nationwide statements that the current repo does not justify.

All reported timings and parity statements in this paper come from the accepted, final unified artifacts in the repository's audited goal history. Earlier provisional or superseded measurements are intentionally excluded here.

C. Experimental Environment

Table I summarizes the host environment for the accepted bounded package. All accepted four-system runs in this paper were executed on the same Linux validation workstation.

D. Packages Used In This Paper

The strongest accepted bounded package in the current repo consists of:

- County $\bowtie$ Zipcode on top4_tx_ca_ny_pa,
- BlockGroup $\bowtie$ WaterBodies on county2300_s10,
- bounded LKAU $\bowtie$ PKAU on sunshine_tiny,
- bounded LKAU $\bowtie$ PKAU overlay-seed analogue.

These packages were validated on the same Linux validation workstation. The bounded LKAU $\bowtie$ PKAU package was staged from live OSM Overpass way geometry and then frozen as the accepted bounded Australia slice [11].

E. Overlay-Seed Contract

RTDL's current overlay result is not a materialized polygon geometry. The accepted overlay row schema is instead an overlay-seed analogue with four columns: left_polygon_id, right_polygon_id, requires_lsi, and requires_pip. This schema records whether a polygon pair requires line-segment intersection refine, point-in-polygon refine, or both, and it is the exact surface compared against PostGIS in the accepted bounded overlay package.

VI. Results

A. External Four-System Closure on Accepted Real-Data Packages

Table II summarizes the strongest accepted real-data packages across PostGIS, the native oracle, Embree, and OptiX.

For all four rows in Table II, the correctness relation is:

$$\text{PostGIS} = \text{native C oracle} = \text{Embree} = \text{OptiX}$$

This is the strongest accepted evidence in the current repo because it combines external database truth with internal multi-backend agreement.

B. Bounded Lakes/Parks Closure

RTDL also closes a bounded lakes/parks analogue on Australia:

The LKAU package is explicitly bounded and derived from live OSM way geometry. It is not continent-scale Australia-family closure. However, it is sufficient to close the first accepted four-system overlay-seed analogue row in the project. The corresponding native-oracle times are 1.9138 s for LSI, 0.0915 s for PIP, and 5.1447 s for the overlay-seed analogue.

C. Scalability Analogues

The RayJoin Section 5.6 style scalability surfaces are closed as bounded analogues:

- RayJoin Figure 13: LSI query time and throughput under synthetic scaling,
- RayJoin Figure 14: PIP query time and throughput under synthetic scaling.

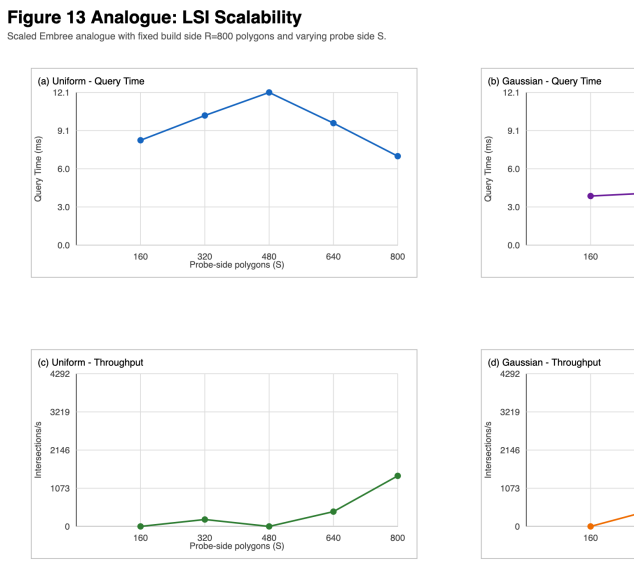
These are accepted as done-bounded, not paper-scale reproductions. They remain valuable because they preserve experiment structure while staying within the project's repeatable bounded execution budget.

In the accepted repo package, these analogues are driven by deterministic synthetic scaling generators rather than the original full RayJoin data volume. They preserve the same experiment shape—query time and throughput under controlled scale-up—while remaining within the validated bounded execution budget of the current project.

TABLE II

Accepted bounded four-system results on the main real-data packages. Reported times are the accepted artifact timings carried forward from the final audited package reports; they are not averages across a new paper-only rerun. The Hits column is applicable only to PIP rows.

Package	Workload	Rows	Hits	PostGIS (s)	Native Oracle (s)	Embree (s)	OptiX (s)
County ⋈ Zipcode top4_tx_ca_ny_pa	LSI	107513	—	34.0323	89.3365	80.2707	53.2900
County ⋈ Zipcode top4_tx_ca_ny_pa	PIP	16352128	7817	0.4300	24.4170	16.8128	19.2551
BlockGroup ⋈ WaterBodies county2300_s10	LSI	216	—	0.2320	0.2286	0.2228	0.8933
BlockGroup ⋈ WaterBodies county2300_s10	PIP	71176	197	0.0084	0.1496	0.1172	0.6477



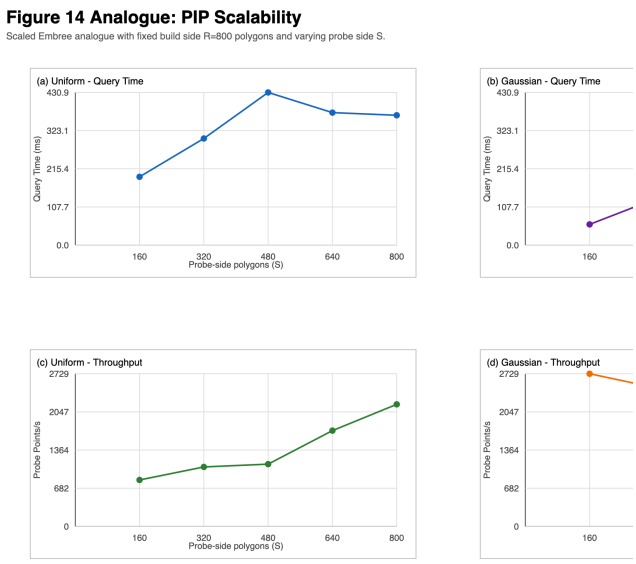


Fig. 1. Accepted bounded analogues of the RayJoin Figure 13 (left, LSI scalability) and Figure 14 (right, PIP scalability) surfaces. These figures are not paper-scale reproductions; they are the accepted bounded synthetic analogues from the RTDL reproduction package.

TABLE III

Accepted bounded LKAU ⋈ PKAU package on sunshine_tiny.

Workload	Rows	PostGIS (s)	Embree (s)	OptiX (s)
LSI	15	0.0623	2.1286	0.5072
PIP	73920 (22 hits)	0.0041	0.0571	0.3844
Overlay-seed	73920	0.2285	0.0850	0.5674

D. Performance Interpretation

Figure 2 summarizes the accepted RayJoin Figure 15-style overlay speedup analogue. As with the rest of the overlay surface in this paper, it is a bounded overlay-seed result rather than a full polygon overlay materialization claim.

The current results do not support a single universal backend winner. OptiX is already beneficial on the largest accepted County/Zipcode package, but Embree remains highly competitive and sometimes faster on smaller slices. PostGIS is extremely fast on indexed positive-hit PIP queries, but the database and RTDL are not executing identical end-to-end work because RTDL materializes full truth rows while PostGIS is measured as an indexed hit query expanded into RTDL equivalent truth for parity.

VII. How RTDL Mirrors the RayJoin Paper

RTDL mirrors RayJoin along three important dimensions and then differs from it in one explicit way.

A. Workload Surface

The paper-facing RTDL slice covers the same workload family categories that matter to RayJoin: LSI, PIP, and overlay composition.

B. Artifact Structure

RTDL now has a bounded paper-facing closure package covering:

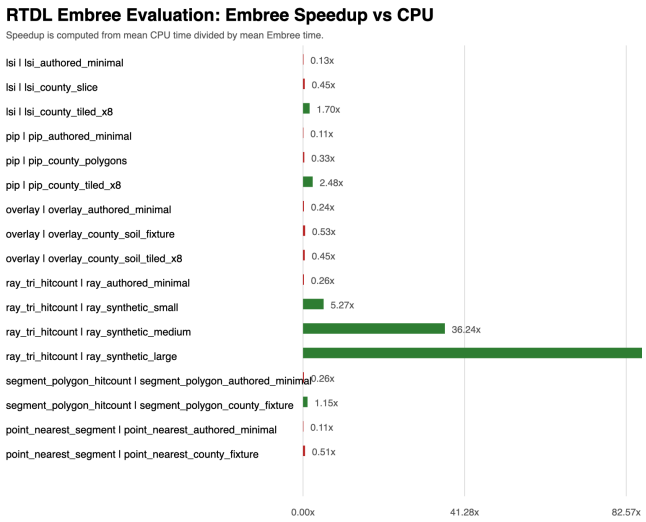


Fig. 2. Accepted bounded Figure 15-style overlay speedup analogue. The plotted surface corresponds to RTDL's current overlay-seed semantics, not full polygon overlay output geometry.

- RayJoin Table 3 style bounded package rows,
- RayJoin Figure 13 bounded scalability analogue (Figure 1, left),
- RayJoin Figure 14 bounded scalability analogue (Figure 1, right),
- RayJoin Table 4 bounded overlay-seed analogue,
- RayJoin Figure 15 bounded overlay-seed speedup analogue (Figure 2).

C. Evaluation Discipline

The project mirrors the paper's experiment structure while replacing weak claims with explicit fidelity labels. Rather than claiming unavailable or unstable dataset families are done, RTDL defers them explicitly. Rather than claiming full overlay materialization, RTDL labels the current closure as an overlay-seed analogue. This produces a more reviewable and reproducible paper story than an overstated "full reproduction" claim would.

D. Key Differences from the Original Paper

The current RTDL paper package differs from the original RayJoin paper in four main ways:

1) It is bounded rather than paper-scale.
2) It relies on accepted analogues where exact paper data is unavailable.
3) It uses PostGIS as an external correctness reference, which RayJoin did not use in the same way.
4) The current overlay surface is a seed-generation analogue, not full polygon overlay materialization.

VIII. Limitations

This work is intentionally explicit about its limitations.

First, this is not a paper-identical reproduction. Several continent-scale lakes/parks families remain deferred because their public acquisition paths are unstable or unavailable. In the current accepted package these deferred families are:

- LKAF $\bowtie$ PKAF,
- LKAS $\bowtie$ PKAS,
- LKEU $\bowtie$ PKEU,
- LKNA $\bowtie$ PKNA,
- LKSA $\bowtie$ PKSA.

Second, the accepted practical analogue for the paper's Block $\bowtie$ Water row is the bounded county2300_s10 package over BlockGroup and WaterBodies. That scoping qualifier matters and must not be erased in downstream summaries.

Third, only bounded LKAU $\bowtie$ PKAU currently has four-system overlay-seed closure. County/Zipcode and BlockGroup/WaterBodies remain useful overlay analogue context from earlier Embree work, but not the same four-system overlay closure.

Fourth, Vulkan is not part of the accepted paper package. It remains provisional.

IX. Conclusion

RTDL now supports a bounded, externally checked, multi-backend reproduction of the RayJoin workload family that is strong enough to describe in paper form. The current contribution is not a claim of full paper-identical reproduction. Instead, it is a paper-facing bounded closure package grounded in accepted evidence: synthetic scalability analogues for RayJoin Figure 13 and Figure 14, bounded package closure for the stable RayJoin Table 3 families, and bounded overlay-seed analogue closure for RayJoin Table 4 and Figure 15. The result is a trustworthy baseline for peer review and subsequent workload expansion, rather than a backend bring-up story or a speculative full-reproduction claim.

Acknowledgment

This manuscript is based on the current audited RTDL repository state and its accepted multi-agent review process.

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